

VISHAL DEEP

Dallas, TX, USA | [LinkedIn](#) | [Google Scholar](#)

RESEARCH INTERESTS

Embedded Systems, Batteryless Devices, Wireless Communication, Intermittent Computing; Internet-of-things (IoT), Wireless Sensor Network (WSN), Time Synchronization, Computer Architecture

EDUCATION

Iowa State University

August 2018 - August 2024

Ph.D., Computer Engineering

GPA: 3.83

- Dissertation title: "Establishing Time in Batteryless Intermittently-powered Sensor Nodes"
- Advisor: Dr. Henry Duwe

California State University - Fresno

August 2014 - December 2016

Master's, Computer Engineering

GPA: 3.88

- Thesis title: "Realization of state-of-the-art intra-prediction in high efficiency video coding (HEVC)"
- Advisor: Dr. Tarek Elarabi

Kurukshetra University

August 2011 - May 2013

Master's, Microelectronics and VLSI Design

GPA: 3.5

Kurukshetra University

August 2006 - May 2011

Bachelor's, Computer Engineering

PROFESSIONAL EXPERIENCE

Lennox International (Product Development & Research)

Dallas, TX, USA

EMBEDDED SOFTWARE ENGINEER

January 2025 - Present

- Led the implementation of twin EBM Direct+ blower and Auto Addressing features of Modbus devices for commercial HVAC units, designing the twin blower system to address DoE efficiency challenges, responsibilities included design documentation, unit testing, and system integration.
- Evaluated and integrated new AI development tools and streamlined firmware build/test processes, resulting in improved team productivity and accelerated adoption to advanced technologies.
- Developed an advanced Refrigerant Charge Assistant algorithm for next-generation S40 smart thermostats, leveraging agile methodologies, and achieving an 80% increase in the automation of charging refrigerant.
- Implemented comprehensive unit tests in the GoogleTest framework and integration tests using a custom testing application, resulting in a 20% reduction in bugs and improved code quality.
- Created detailed project documentation, including UML diagrams such as class diagrams, state machine diagrams, and sequence diagrams, resulting in a 20% increase in efficiency during the development phase.
- Participated in personal skill development events, "Coffee Chat with Leaders" and "Lunch & Learn Series," resulting in an improvement in communication, leadership, and teamwork skills.

Iowa State University

Ames, IA, USA

RESEARCH ASSISTANT

August 2018 – August 2024

- Led the team in implementing novel time synchronization mechanism (LEASST) and error metrics for batteryless nodes using a Python simulator, leading to a reduction in error by 9x on a network of 50 nodes.
- Enhanced the computation and communication capabilities of batteryless devices by integrating MSP430 MCU and CC1352R Sensor Tag with ARM Cortex M4F processor, resulting in a 30% increase in communication throughput.
- Designed and prototyped an accurate persistent clock using an MSP430 MCU, ensuring timekeeping accuracy within 10% even during extended power-off periods, significantly advancing time measurement in powered-off devices.
- Implemented a data logging system using Raspberry Pi 4 and UART communication, reducing time spent on data collection and analysis by 60%, optimizing data science workflows with Python.
- Mentored and coached undergraduate student teams in developing innovative solutions for their senior design projects, resulting in 50% of projects being selected for presentations at prestigious industry conferences.

DPS Telecom

Fresno, CA, USA

EMBEDDED SYSTEM PROGRAMMER

February 2017 - August 2017

- Developed and implemented remote monitoring firmware utilizing C/C++ for PIC32 MCU, resulting in a 15% reduction in device downtime and a 20% increase in overall system reliability.
- Prototyped antenna rotation system for EMI testing using Arduino and stepper motors, reducing testing time by 40%.
- Troubleshooted and resolved over 80 firmware bugs within 6 months, utilizing oscilloscopes and logic analyzers for testing, and debugging, resulting in a 40% reduction in device malfunctions and improving overall system stability.

- Conducted thorough analysis of operating temperature and EMI radiation tests on manufactured devices, resulting in a 90% decrease in product defects and a significant improvement in product quality metrics.
- Executed regular testing of SOP to ensure accuracy and effectiveness, leading to a 10% improvement in team collaboration and knowledge transfer among members.

California State University - Fresno
RESEARCH/TEACHING ASSISTANT

Fresno, CA, USA
January 2015 - December 2016

- Enhanced GPS receiver accuracy by utilizing embedded sensor data from smartphones in MATLAB within a collaborative team. Developed predictive algorithms using IMU sensors to forecast paths in GPS signal-challenged areas.
- Supervised students in the Computer Architecture lab, providing guidance on Verilog assignments and projects.
- Managed grading of assignments, quizzes, and exams while creating educational materials for undergraduate classes.

HONORS and AWARDS

- Selected and funded to present at the prestigious ACM SIGBED Student Research Competition during ESWEEK 2023 in Germany, showcasing excellence in embedded systems research.
- Recognized as the runner-up for the best PhD forum presentation at the 20th ACM Conference on Embedded Networked Sensor Systems (SenSys 2022), showcasing excellent communication and research skills.
- Received the graduate Dean's Medal for the class of 2016, highlighting outstanding academic achievements and leadership abilities.
- Honored as an exceptional student at Lyles College of Engineering (LCOE) in 2016, demonstrating a commitment to engineering excellence.
- Recipient of the prestigious Charles E. Buckley Engineering Scholarship at CSU Fresno, acknowledging exceptional academic and leadership skills.
- Awarded the esteemed Dean's Scholarship by Lyles College of Engineering, reflecting outstanding academic performance and potential in engineering.
- Secured the highly competitive GATE scholarship from the Ministry of Human Resource Development (MHRD), Government of India, for exemplary performance in engineering and a commitment to academic excellence.

PUBLICATIONS

1. V. Narayanan, M. Gshash, **Vishal Deep**, M. Wymore, D. Qiao, Neihart, N. M., & H. Duwe, (2025, March). PAIL: Predictable and Adaptive Intermittent Lifecycling for Robust Coordination Between Batteryless Systems. In *2025 IEEE International Conference on Pervasive Computing and Communications (PerCom)* (pp. 138-144). IEEE.
2. Rohit Sahu, **Vishal Deep**, and Henry Duwe. "HANNA: Harvesting-Aware Neural Network Architecture Search for Batteryless Intermittent Devices" In 2024 IEEE 43rd International Performance Computing and Communications Conference (IPCCC).
3. Wymore, Mathew., Rohit Sahu, Thomas Ruminski, **Vishal Deep**, Morgan Ambourn, Gregory Ling, Vishak Narayanan, William Asiedu, Daji Qiao, and Henry Duwe. "Lure: A simulator for networks of batteryless intermittent nodes." *Performance Evaluation* 166 (2024): 102440.
4. S. Fu, V. Narayanan, M. L. Wymore, **Vishal Deep**, H. Duwe and D. Qiao, "No battery, no problem: Challenges and opportunities in batteryless intermittent networks," in *Journal of Communications and Networks*, vol. 25, no. 6, pp. 806-813, Dec. 2023
5. **Vishal Deep**, Wymore, Mathew L., Daji Qiao, and Henry Duwe. "Toward a Shared Sense of Time for a Network of Batteryless, Intermittently-powered Nodes" In 2022 IEEE 41st International Performance Computing and Communications Conference (IPCCC).
6. **Vishal Deep**, Mathew L. Wymore, Alexis A. Aurandt, Vishak Narayanan, Shen Fu, Henry Duwe, and Daji Qiao. "Experimental Study of Lifecycle Management Protocols for Batteryless Intermittent Communication." In 2021 IEEE 18th International Conference on Mobile Ad Hoc and Smart Systems (MASS), pp. 355-363. IEEE, 2021.
7. **Vishal Deep**, Vishak Narayanan, Mathew Wymore, Daji Qiao, and Henry Duwe. "HARC: A heterogeneous array of redundant persistent clocks for batteryless, intermittently-powered systems." In 2020 IEEE Real-Time Systems Symposium (RTSS), pp. 270-282. IEEE, 2020.
8. Wymore, Mathew L., **Vishal Deep**, Vishak Narayanan, Henry Duwe, and Daji Qiao. "Lifecycle management protocols for batteryless, intermittent sensor nodes." In 2020 IEEE 39th International Performance Computing and Communications Conference (IPCCC), pp. 1-8. IEEE, 2020.
9. **Vishal Deep**, Aditi Mishra, Daji Qiao, and Henry Duwe. "Revisiting time remanence clocks for energy harvesting wireless sensor nodes." In *Proceedings of the 7th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems*, pp. 58-59. 2019.
10. **Vishal Deep**, and Tarek Elarabi. "Efficient IEEE 802.15. 4 ZigBee standard hardware design for IoT applications." In 2017 International Conference on Signals and Systems (ICSigSys), pp. 261-265. IEEE, 2017.
11. **Vishal Deep**, and Tarek Elarabi. "HEVC/H. 265 vs. VP9 state-of-the-art video coding comparison for HD and UHD applications." In 2017 IEEE 30th Canadian Conference on Electrical and Computer Engineering (CCECE), pp. 1-4. IEEE, 2017.
12. **Vishal Deep**, and Tarek Elarabi. "Write latency reduction techniques of state-of-the-art phase change memory." In 2016 European Modelling Symposium (EMS), pp. 213-217. IEEE, 2016.

13. Abhijit Suprem, **Vishal Deep**, and Tarek Elarabi. "Orientation and displacement detection for smartphone device based IMUs." IEEE Access 5 (2016): 987-997.
14. Tarek Elarabi, Bhanu Sharma, Karan Pahwa, and **Vishal Deep**. "Big data analytics concepts and management techniques." In 2016 International Conference on Inventive Computation Technologies (ICICT), vol. 2, pp. 1-6. IEEE, 2016.
15. **Vishal Deep**. Realization of state-of-the-art intra-prediction in High Efficiency Video Coding (HEVC). California State University, Fresno, 2016.
16. Tarek Elarabi, **Vishal Deep**, and Chashamdeep Kaur Rai. "Design and simulation of state-of-art ZigBee transmitter for IoT wireless devices." In 2015 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), pp. 297-300. IEEE, 2015.

CONFERENCE PRESENTATIONS

- 2023 ACM SIGBED Student Research Competition (SRC) in Hamburg, Germany.
- 2022 ACM 20th Conference on Embedded Networked Sensor Systems (SenSys) in Boston, MA (USA).
- 2022 IEEE 41st International Performance, Computing, and Communications Conference (IPCCC) in Austin, TX (USA).
- 2021 IEEE 18th International Conference on Mobile Ad Hoc and Smart Systems (MASS).
- 2020 IEEE 41st Real-Time Systems Symposium (RTSS).
- 2019 ACM Proceedings of the 7th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (ENSSys) in New York City, NY (USA).
- 2017 IEEE International Conference on Signals and Systems (ICSigSys) in Bali (Indonesia).
- 2015 IEEE 15th Symposium on Signal Processing and Information Technology (ISSPIT).

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Embedded C, Python, MATLAB, Simulink, Bash, Verilog
- **Embedded Development:** FPGA, UART, SPI, I2C, ADC, GPIO, RTOS
- **Hardware Debugging:** Oscilloscope, Logic Analyzer, Protocol Analyzer, Spectrum Analyzer
- **Software Tools:** Code Composer Studio, Xilinx Vivado, Altera Quartus II, ModelSim, Contiki OS, TI RTOS, Visual Studio Code, Visual Studio, Accurev, Collaborator, WSL, Git, CMake, GDB, Hardware Simulation
- **Machine Learning:** NumPy, Matplotlib, Jupyter, Tensorflow, Scikit-learn, Keras, Pandas
- **Web Development:** Angular, NodeJS, HTML/CSS, JavaScript, MongoDB, SQL

PROFESSIONAL SERVICES

- Reviewed research article for IEEE International Conference on Computer Communications (INFOCOM 2024, 2025).
- Judged innovative engineering projects at the 65th State Science & Technology Fair of Iowa in 2022.
- Reviewed high-impact research manuscripts for prestigious journals such as IEEE Access and Advances in Science, Technology and Engineering Systems Journal (ASTESJ).
- Reviewed research article for International Conference on Software Defined Systems (SDS) in 2020.
- Contributed as a Program Committee member for the ACM Urban Sensor Network workshop, jointly hosted with SenSys.
- Served as a member of organizing committee of NASET 2012 (a national-level symposium in India).

PROFESSIONAL MEMBERSHIPS

- IEEE Member
- ACM Member
- ACM SIGBED (Special Interest Group on Embedded Systems) Member

MOOCS and CERTIFICATES

- VLSI CAD Logic by University of Illinois, Urbana-Champaign, August 2017, Coursera License -2U72Z8PYEJPB
- An Introduction to Interactive Programming in Python by Rice University, February 2017, Coursera license - CTV6Y8XA5CP7
- Object-Oriented Programming with Java, University of Helsinki, October 2016
- Fundamentals of Digital Image and Video Processing, Northwestern University, May 2016 Coursera License - DP9SAWJ96EU8
- Embedded System Programming, Center for Development and Advanced Studies, India, August 2009, License - 09-10/ST-ESP/5149

REFERENCES

Available on request